

have prior information about the location of well suited bandwidths, we recommend using the proposed collection of bandwidths with uniform weights as defined in Section 5.1. We use $B_1 = 2000$ and $B_2 = 2000$ simulated test statistics to estimate the quantiles and the probability in Equation (13) for the level correction, respectively, and use $B_3 = 50$ steps of bisection method. We refer to this test as **MMDagg*** and emphasize the fact that it is run with exactly the same parameters across all experiments.

5.3 State-of-the-art MMD-based two-sample tests

Gretton et al. (2012a) first suggested using the median heuristic to choose the bandwidth of the MMD test with the Gaussian kernel⁸ corresponding to $K_i(u) := \frac{1}{\sqrt{\pi}} \exp(-u^2)$ for $u \in \mathbb{R}$, $i = 1, \dots, d$, so that

$$k_\lambda(x, y) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{x_i - y_i}{\lambda_i}\right) = \frac{1}{\lambda_1 \dots \lambda_d \pi^{d/2}} \exp\left(-\sum_{i=1}^d \left(\frac{x_i - y_i}{\lambda_i}\right)^2\right).$$

They proposed to set the bandwidth to be equal to

$$\lambda_i := \text{median}\{\|w - w'\|_2 : w, w' \in \mathbb{X}_m \cup \mathbb{Y}_n, w \neq w'\}$$

for $i = 1, \dots, d$. To generalise this approach to our case where $K_1, \dots, K_d$ need not all be the same, as explained in Section 5.1, we can in a similar way set the bandwidth⁹ to

$$\lambda_i := \text{median}\{|w_i - w'_i| : w, w' \in \mathbb{X}_m \cup \mathbb{Y}_n, w \neq w'\}$$

for $i = 1, \dots, d$. With this specific definition for the bandwidth, we use the notation $\lambda_{med} := (\lambda_1, \dots, \lambda_d)$. We refer to the single MMD test with the bandwidth λ_{med} as **median** in our experiments.

Another common approach for selecting the bandwidth was first introduced by Gretton et al. (2012b) for the single MMD test using the linear-time MMD estimator. The method was then extended to the case of the quadratic-time MMD estimator by Sutherland et al. (2017). It consists in splitting the data in two parts and in using the first part to select the bandwidth which maximises the asymptotic power of test, or equivalently the estimated ratio (Liu et al., 2020, Equations 4 and 5)

$$\frac{\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_n, \mathbb{Y}_n)}{\widehat{\sigma}_\lambda(\mathbb{X}_n, \mathbb{Y}_n)} \quad (16)$$

where

$$\widehat{\sigma}_\lambda^2(\mathbb{X}_n, \mathbb{Y}_n) := \frac{4}{n^3} \sum_{i=1}^n \left(\sum_{j=1}^n h_\lambda(X_i, X_j, Y_i, Y_j) \right)^2 - \frac{4}{n^4} \left(\sum_{i=1}^n \sum_{j=1}^n h_\lambda(X_i, X_j, Y_i, Y_j) \right)^2 + 10^{-8}$$

8. Gretton et al. (2012a) actually consider the unnormalised Gaussian kernel without the $(\lambda_1 \dots \lambda_d \pi^{d/2})^{-1}$ term, but as pointed out in Footnote 1, this does not affect the output of the test.

9. Note that those two ways of setting the bandwidths are not equivalent for the Gaussian kernel but they are each equally valid.

is a regularised positive estimator of the asymptotic variance of the quadratic-time estimator $\widehat{\text{MMD}}_{\lambda}^2(\mathbb{X}_n, \mathbb{Y}_n)$ under the alternative hypothesis $\mathcal{H}_a$ for $m = n$. In our experiments, we select the bandwidth of the form $c\lambda_{med}$ for various positive values of c which maximises the estimated ratio. The single MMD test with the selected bandwidth is then performed on the second part of the data. In our experiments, we refer to this test as **split**.

Another interesting test to compare ours to, is the one which uses new data to choose the optimal bandwidth. This corresponds to running the above test which uses data splitting on twice the amount of data. In some sense, this represents the best performance the single MMD test can achieve as the test is run on the whole dataset with an optimal choice of bandwidth. As such, it is interesting to observe the difference in power between MMDAgg and this oracle test which uses extra data. In our experiments, we denote this test as **oracle**.

A radically different approach to constructing an MMD adaptive two-sample test was recently presented by Kübler et al. (2020). They work in the asymptotic regime and require asymptotic normality under the null hypothesis of their MMD estimator, so they are restricted to using the linear-time estimator. Using all of the data, they compute the linear-time MMD estimates for several kernels (or several bandwidths of a kernel), they then select the linear combination of these which maximises a proxy for asymptotic power, and compare its value to their test threshold. They do not split the data but they are able to correctly calibrate their test for the introduced dependencies. For this, they prove and use a generalised version of the post-selection inference framework (Fithian et al., 2014; Lee et al., 2016) which holds for uncountable candidate sets (i.e. all linear combinations). In our experiments, we compare our aggregated MMDAgg test to their one-sided test (OST—Kübler et al., 2020) for which we use their implementation. This test is referred to as **ost** in our experiments.

In later work, Kübler et al. (2022b) propose an AutoML (Automated Machine Learning) test with an implementation which is essentially parameter-free. Their test relies on sample splitting, on cross-validation, and on permuting the data, the witness function of the test is learnt by training an AutoGluon model (Erickson et al., 2020) for some prescribed amount of time (one minute by default). Depending on the time limit and on the compute available, a different model will be chosen automatically. While such a black-box approach can certainly be convenient for everyday users, this convenience comes at the expense of reproducibility (even on the same machine it can take different amounts of time to train identical models). We refer to this test in our experiments as **AutoML**.

5.4 Experimental procedure

To compute the median bandwidth

$$(\lambda_{med})_i := \text{median}\{|w_i - w'_i| : w, w' \in \mathbb{X}_m \cup \mathbb{Y}_n, w \neq w'\}$$

for $i = 1, \dots, d$, we use at most 1000 randomly selected data points from $\mathbb{X}_m$ and at most 1000 from $\mathbb{Y}_n$, since the median is robust, this is sufficient to get an accurate estimate of it. Moreover, we use a threshold so that the bandwidth is not smaller than 0.0001. This avoids division by 0 in some settings where one component of the data points is always the

same value, as it can be the case for the problem considered in Section 5.6 which uses the MNIST dataset, where the pixel in one corner of the images is always black for every digit.

We run all our experiments with the Gaussian kernel

$$k_\lambda(x, y) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{x_i - y_i}{\lambda_i}\right) = \frac{1}{\lambda_1 \cdots \lambda_d \pi^{d/2}} \exp\left(-\sum_{i=1}^d \left(\frac{x_i - y_i}{\lambda_i}\right)^2\right)$$

for $K_i(u) := \frac{1}{\sqrt{\pi}} \exp(-u^2)$ for $u \in \mathbb{R}$, $i = 1, \dots, d$, and with the Laplace kernel

$$k_\lambda(x, y) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{x_i - y_i}{\lambda_i}\right) = \frac{1}{\lambda_1 \cdots \lambda_d 2^d} \exp\left(-\sum_{i=1}^d \left|\frac{x_i - y_i}{\lambda_i}\right|\right)$$

for $K_i(u) := \frac{1}{2} \exp(-|u|)$ for $u \in \mathbb{R}$, $i = 1, \dots, d$. As mentioned in Footnote 1, MMDAgg does not depend on the scaling of the kernels. Hence, in our implementation we drop the scaling terms in front of the exponential functions, which is numerically more stable.

We use a wild bootstrap for all our experiments, except for the one in Appendix A.3 where we compare using the permutation-based and wild bootstrap procedures, and for the one in Appendix A.4 where we must use permutations as we consider different sample sizes $m \neq n$. We use level $\alpha = 0.05$ for all our experiments.

We run all our tests on three different types of data: 1-dimensional and 2-dimensional perturbed uniform distributions, and the MNIST dataset. Those are introduced in Sections 5.5 and 5.6, respectively. We use sample sizes $m = n = 500$ for the 1-dimensional perturbed uniform distributions and for the MNIST dataset, and use larger sample sizes $m = n = 2000$ for the case of the 2-dimensional perturbed uniform distributions.

For the **split** and **oracle** tests, we use two equal halves of the data, and **oracle** is run on twice the sample sizes. We choose the bandwidth which maximises the estimated ratio presented in Equation (16) out of the collection $\{c\lambda_{med} : c \in \{0.1, 0.2, \dots, 0.9, 1\}\}$ when considering perturbed uniform distributions, and when considering the MNIST dataset we select it out of the collection $\{2^c \lambda_{med} : c \in \{10, 11, \dots, 19, 20\}\}$. Similarly to our aggregated tests of Section 5.1, for the **median**, **split** and **oracle** tests, we use $B = 500$ simulated test statistics to estimate the quantile.

For **MMDAgg***, we also use either the Gaussian or the Laplace kernel with $N = 10$ bandwidths, we refer to those tests as **MMDAgg* Gaussian** and **MMDAgg* Laplace**. We also consider aggregating over different types of kernels, in particular, we can use both the Gaussian and Laplace kernels, each with $N = 10$ bandwidths. This gives a collection consisting of $2N = 20$ kernels, over which **MMDAgg* Laplace Gaussian** aggregates. Finally, we propose to aggregate 12 kernels (each with $N = 10$ bandwidths): the Gaussian kernel, the inverse multiquadric (IMQ) kernel, and the Matérn kernels with the ℓ^1 and ℓ^2 distances for $\nu = 0.5, 1.5, 2.5, 3.5, 4.5$ (the Laplace kernel is the Matérn kernel with ℓ^1 distance and $\nu = 0.5$). This test aggregates over $12N = 120$ kernels, we refer to it as **MMDAgg* A11**. Note that in Figure 6, we consider $N = 1000$ bandwidths, which means for example that the **MMDAgg* A11** aggregates over $12N = 12000$ kernels, and we observe that it retains its high power.

To estimate the power in our experiments, we average the test outputs of 500 repetitions, that is, 500 times, we sample some new data and run the test. We sample new data for each